

6.2 $V = \text{Span} \{ (2, 1, 0, 1)^T, (1, 0, 1, 2)^T \}$

$$u_1 = (2, 1, 0, 1)^T \quad u_2 = (1, 0, 1, 2)^T$$

$$f(u_1) = (4, 1, 2, 5)^T \quad f(u_2) = (2, 2, -2, -2)^T$$

→ obratny káž v V lebo:

$$f(u_1) = 1 \cdot u_1 + 2 \cdot u_2 = (2, 1, 0, 1)^T + (2, 0, 2, 4)^T = (4, 1, 2, 5)^T$$

$$f(u_2) = 2 \cdot u_1 - 2 \cdot u_2 = (4, 2, 0, 2)^T - (2, 0, 2, 4)^T = (2, 2, -2, -2)^T$$

Matica operátora f vzhľadom k báze $B = (u_1, u_2)$ je

$$[f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

Dalej musíme vypočítať v. čísla pomocou char. polynómu.

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda+3)(\lambda-2)$$

v. čísla sú $\lambda_1 = 2, \lambda_2 = -3$

v. vektory vzhľadom k báze B $[v]_B$ sú:

$$\lambda_1 = 2: \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \quad o_2 = + \quad o_1 = 2+ \quad o_2 = 1 \quad \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} = x_1$$

$$\lambda_2 = -3: \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \quad o_4 = + \quad o_3 = -\frac{2}{2} \quad o_3 = -2 \quad \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\} = x_2$$

výjadriť vektor $(5, 1, 3, 7)^T$ v báze B

$$(5, 1, 3, 7)^T = a(2, 1, 0, 1)^T + b(1, 0, 1, 2)^T$$

$$a = 1 \quad b = 3 \quad [w]_B = (1, 3)^T$$

Pre $f^n(w)$ máme $(1, 3)^T = c_1(2, 1)^T + c_2(1, -2)^T$

$$2c_1 + c_2 = 1$$

$$c_1 - 2c_2 = 3 \Rightarrow c_1 = 3 + 2c_2$$

$$2(3 + 2c_2) + c_2 = 1 \Rightarrow c_2 = -1, c_1 = 1 \quad \text{teda } [w]_B = 1 \cdot x_1 - 1 \cdot x_2$$

aplikujeme f^n

$$[f^n(w)]_B = 1 \cdot 2^n \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 1 \cdot (-3)^n \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2^n - (-3)^n \\ 2^n + 2 \cdot (-3)^n \end{pmatrix}$$

teda výsledok prepíšeme ako lin. kombináciu u_1 a u_2

$$f^n((5, 1, 3, 7)^T) = (2^{n+1} - (-3)^n) \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (2^n + 2 \cdot (-3)^n) \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$